

Training: physics-informed Gaussian process

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Objectives

The objectives of the training session are threefold

- To discover **Gaussian process** (GP) and **kernel methods**,
- To understand how they can handle 'physical' information, in particular the one brought by a **partial differential equation** (PDE)
- To **customize yourself a GP** and work on simple examples.

Pedagogical means

- Synthetic resources: **textbook** + slides
- Practical activities: typical exercises, computer labs

Why Gaussian processes?

GPs are a good complement to deep-learning methods, and widely used in machine learning/uncertainty quantification (UQ). The reasons are many:

- They work with scarce data (and can be adapted to handle huge datasets)
- They provide uncertainty quantification (confidence/credibility intervals)
- They are suitable for approximation problems (they extend splines)
- They are parameterized by two functions (trend, kernel)
 - ▶ highly flexible, they can be defined on general spaces
- They naturally handle linear constraints (e.g. derivatives)
- They are mathematically sounded
 - ▶ at the crossroad between three well-established theories: random processes, reproducing kernel Hilbert spaces (RKHS), positive definite functions.
- ...

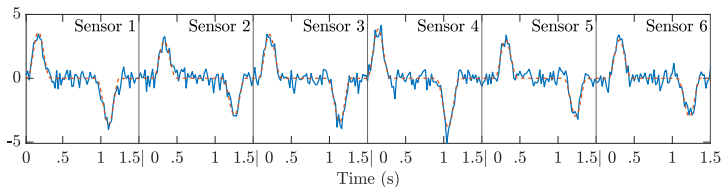
Some applications w.r.t. PDEs

In general, GP models are relevant for **surrogate modelling** of expensive computer codes → sensitivity analysis, UQ, ...

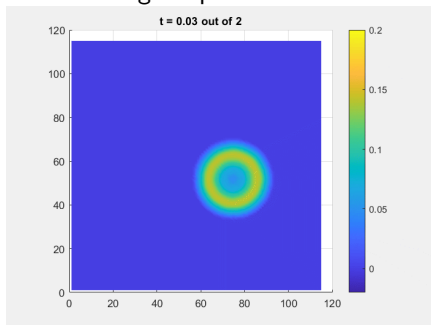
In the PDE setting, GPs are also relevant for **inverse problems** :

- Efficient **estimation of the solution of a PDE** given scattered pointwise measurements of the solution;
- **Estimation of the physical parameters of the PDE** by calibrating the hyperparameters of the kernel method;
- **Better uncertainty quantification** on the estimated sol. of a PDE ;

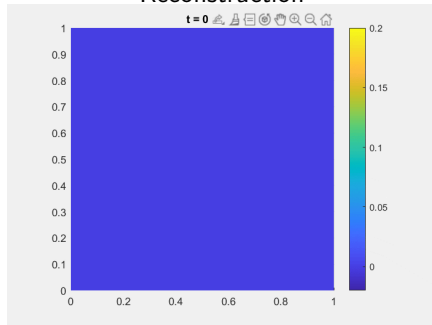
Ex 1 - Estimation of the solution of the wave equation



Original phenomenon



Reconstruction



Ex 2 - Better uncertainty quantification : 1D heat equation

Estimate the sol. of $\partial_t u = D \partial_{xx}^2 u$, $u(0, x) = u_0(x)$ where u_0 is unknown.

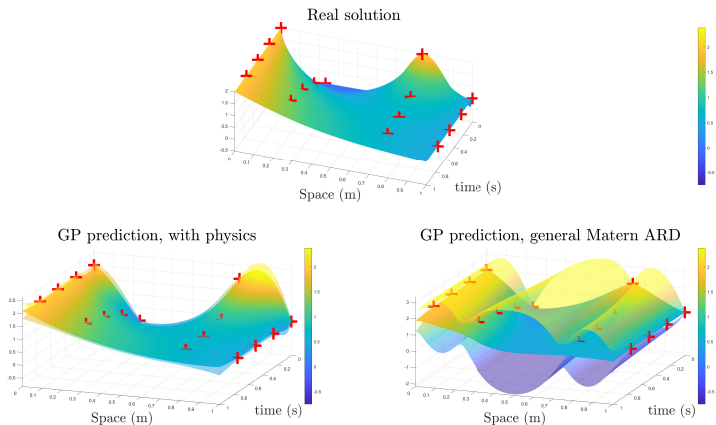


Figure: Red : measurement points. The prediction (and any realization) of the physics-informed GP verifies the equation **everywhere**.

Ex 3 - Solving of the (non-linear) Burgers's equation

Estimate the solution of $\partial_t u + \alpha(x)u\partial_x u - \nu(x)\partial_x^2 u = 0$ on $\Omega = (-1, 1) \times (0, 1]$, with $u(0, x) = -\sin(\pi x)$ and $u(t, -1) = u(t, 1) = 0$.

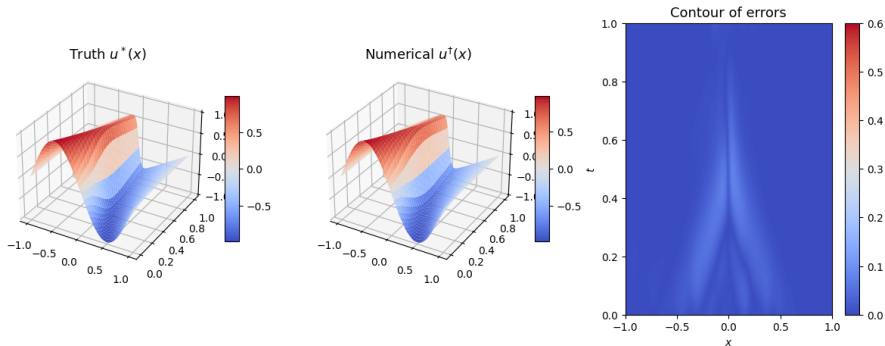


Figure: The estimated solution verifies the equation on a finite set of 'collocation' points (1000 in Ω , 201 in $\partial\Omega$) uniformly sampled. No measurement points have been used (but could be added to improve accuracy).

Program

- Part 0: General introduction or the benefits of using GPs for PDEs
- Part 1: Generalities on GPs. Function approximation (GP regression).

Lab GPs under ~~symmetry constraints~~ (part 1).

- Part 2: GPs subject to linear constraints. Application to linear PDEs.

Lab GPs under symmetry constraints (part 2).

- Part 3: RKHS, function approximation and GP regression.

Lab GPs regression with derivative information.

- Part 4: Solving non-linear PDEs with kernel methods.

Lab Example: Burger's equation.

Outline

1 Introduction

2 Gaussian processes (GP) and GP regression

- Random process, Gaussian process
- Gaussian process regression

3 GPs and linear constraints. Application to linear PDEs

4 RKHS

5 Physics-informed GP : non-linear PDEs

Random processes

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which all the (real-valued) random variables will be defined. We denote by $L^2(\mathbb{P})$ the Hilbert space of square integrable random variables (defined on Ω).

For a given set $\mathbb{X}$, a *random process* (RP) is a family of random variables $Y(x) : \Omega \rightarrow \mathbb{R}$, indexed by $x \in \mathbb{X}$.

We will denote $Y := (Y(x))_{x \in \mathbb{X}}$.

Trajectory, realization or sample path

Let Y be a RP. For a fixed $w \in \Omega$, a *trajectory or realization or sample path* of Y is the function $x \mapsto Y(x)(w)$.

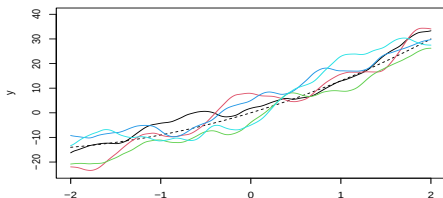


Figure: Five sample paths of a Gaussian process on $\mathbb{X} = [-2, 2]$

Second-order random process, mean, kernel.

Y is a *second-order RP* when all the r.v. $Y(x)$ belong to $L^2(\mathbb{P})$.

By Cauchy-Schwartz inequality, this implies that first moments (expectation) as well as second moments (covariances) are well-defined.

- The *mean* of Y is the function $x \in \mathbb{X} \mapsto \mathbb{E}(Y(x))$.
- The *covariance function* or *kernel* of Y is the function $(x, x') \in \mathbb{X} \times \mathbb{X} \mapsto \text{Cov}(Y(x), Y(x'))$.

Similarly, the *variance* of Y denotes the function $x \in \mathbb{X} \mapsto k(x, x) = \text{Var}(Y(x))$.

Stationarity

- Y is *strongly stationary* if for all locations $x_1, \dots, x_n \in \mathbb{X}$, the law of $(Y(x_1 + t), \dots, Y(x_n + t))$ does not depend on the translation t .

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- Y is *weakly stationary* if for all locations $x_1, \dots, x_n \in \mathbb{X}$, the first two moments of the law of $(Y(x_1 + t), \dots, Y(x_n + t))$ do not depend on t .
Equivalently:

$$\mathbb{E}(Y(x)) = m, \quad k(x, x') = c(x - x')$$

with $m = \mathbb{E}(Y(x_0))$ (for some $x_0 \in \mathbb{X}$) and $c(h) = k(0, h) = k(0, -h)$.

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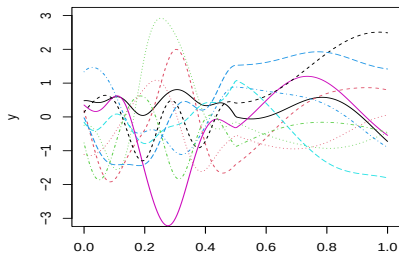
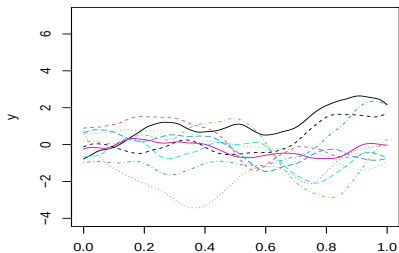
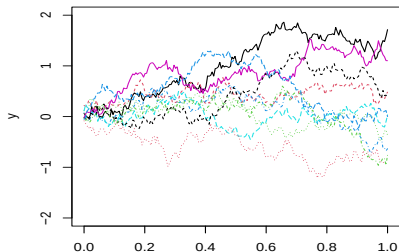
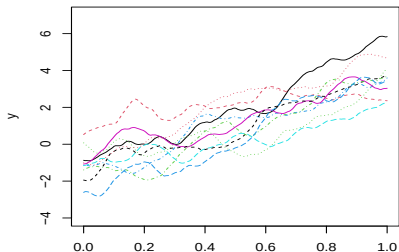
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Obviously, strong stationarity implies weak stationarity.

Exercise

Only one graph corresponds to a stationary process. Which one?

For the others, why the corresponding random process is non stationary?



Gaussian processes

A random process Y defined on $\mathbb{X}$ is a *Gaussian process* (GP) if for all locations $x_1, \dots, x_n \in \mathbb{X}$, the random vector $(Y(x_1), \dots, Y(x_n))$ is a Gaussian vector.

The law of such random vectors is fully characterized by the mean m and the kernel k of Y . We will denote $Y \sim GP(m, k)$.

Reminder on Gaussian vectors

$X := (X_1, \dots, X_d)^\top$ is a Gaussian vector iff it is the affine transformation of independent standard Normal random variables: $\exists \mu \in \mathbb{R}^d, A \in \mathbb{R}^{d \times m}$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)^\top$ where $\varepsilon_1, \dots, \varepsilon_m$ are i.i.d. $\mathcal{N}(0, 1)$, such that

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If $\Gamma := \text{Cov}(X)$ is invertible, X is called non degenerated.

We denote $X \sim \mathcal{N}(\mu, \Gamma)$.

Density function of the multivariate normal distribution

If $X \sim \mathcal{N}(\mu, \Gamma)$ is a non degenerated Gaussian vector in $\mathbb{R}^d$, then X admits the density function

$$f_X(x) = \frac{1}{(2\pi)^{d/2} |\Gamma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Gamma^{-1} (x - \mu) \right)$$

where $|\Gamma| = \det(\Gamma)$.

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This comes directly from the definition, using the theorem of change of variables.

The level sets of the density function (the sets of $x \in \mathbb{R}^d$ such that $f_X(x) = y$, for a given y) are ellipsoids centered at μ , whose axis are given by the eigenvectors of Γ .

Stability by linear mapping

A linear mapping of a Gaussian vector is a Gaussian vector. More precisely, if $X \sim \mathcal{N}(\mu, \Gamma)$ is Gaussian vector on $\mathbb{R}^d$, and $L : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ is a $d \times d'$ matrix, then LX is a Gaussian vector on $\mathbb{R}^{d'}$ with $LX \sim \mathcal{N}(L\mu, L\Gamma L^\top)$

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The result is obvious from the definition: if $X = \mu + A\varepsilon$, then $LX = L\mu + LA\varepsilon$.

Conditioning of Gaussian vectors

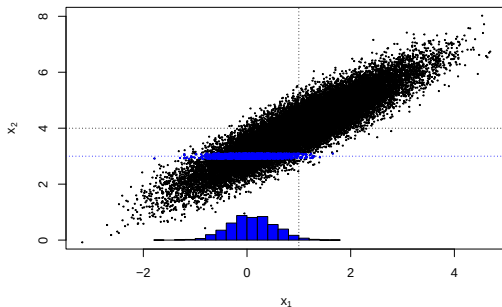


Figure: Illustration of the conditioning of Gaussian vectors on a simulated sample.

Let $U = (V, W) \sim \mathcal{N}(\mu, \Gamma)$ be a Gaussian vector on $\mathbb{R}^d$, where V, W are subvectors of dimension d_V, d_W respectively. Write $\mu = (\mu_V, \mu_W)^\top$ with $\mu_V = \mathbb{E}(V), \mu_W = \mathbb{E}(W)$ and

$$\Gamma = \begin{bmatrix} \Gamma_V & \Gamma_{V,W} \\ \Gamma_{W,V} & \Gamma_W \end{bmatrix}$$

where $\Gamma_V = \text{Cov}(V), \Gamma_W = \text{Cov}(W)$, and $\Gamma_{V,W} = \text{Cov}(V, W) := \mathbb{E}[(V - \mu_V)(W - \mu_W)^\top]$ (similar definition for $\Gamma_{W,V}$).

Then $V|W = w$ is a Gaussian vector on $\mathbb{R}^{d_W}$ with mean and covariance matrix

$$\begin{aligned} \mathbb{E}(V|W = w) &= \mu_V + \Gamma_{V,W}\Gamma_W^{-1}(w - \mu_W) && \text{linear with respect to } w \\ \text{Cov}(V|W = w) &= \Gamma_V - \Gamma_{V,W}\Gamma_W^{-1}\Gamma_{W,V} && \text{does not depend on } w \end{aligned}$$

Some properties of GPs inherited from Gaussian vectors

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- **a GP Y conditioned on $Y(x_i) = y_i, (i = 1, \dots, n)$ is still a GP.**

This is the basis of Gaussian process regression, developed in Chapter 4. By stability of GPs under linearity, this property is true for linear equality constraints (and not only interpolation ones).

Simulation of a GP

A simulation of $Y \sim GP(m, k)$ is possible on a set of discrete locations $X = \{x_1, \dots, x_n\}$.

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A simulation of $Y \sim GP(m, k)$ is possible on a set of discrete locations $X = \{x_1, \dots, x_n\}$. Indeed, then the law of $(Y(x_1), \dots, Y(x_n))^T$ is $\mathcal{N}(m(X), k(X, X))$ where

- $m(X)$ is the vector of size n whose component i is equal to $m(x_i)$
- $k(X, X)$ is the matrix of size n whose coefficient (i, j) is equal to $k(x_i, x_j)$

Obtaining a realization of Y at X , it is thus equivalent to simulating from $\mathcal{N}(m(X), k(X, X))$.

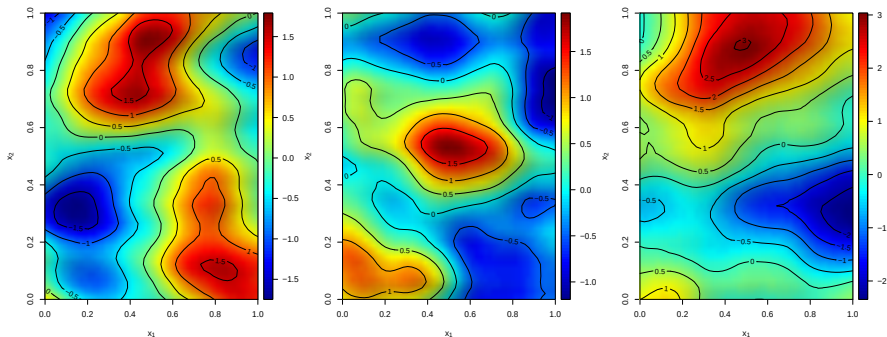


Figure: Simulations of a Gaussian process on $\mathbb{X} = [0, 1]^2$ (obtained by simulating a Gaussian vector on a fine grid of $\mathbb{X}$), with kernel $k(x, x'; \ell) = k_1(x_1, x'_1; 2\ell)k_2(x_2, x'_2; \ell)$, where k_1 is a one-dimensional Matérn 5/2 kernel (see Chap. 3) and ℓ is a parameter.

A functional interpolation problem

Given a set of observations $(x_i, y_i), i = 1, \dots, n$ with $x_i \in \mathbb{X}$ and $y_i \in \mathbb{R}$, we search for an **interpolator** i.e. a function $f : \mathbb{X} \rightarrow \mathbb{R}$ such that

$$f(x_i) = y_i, \quad \text{for all } i = 1, \dots, n$$

This is an **ill-posed problem**, so we need more assumptions.

GP regression¹

Here we view f as one realization of a GP Y with mean m and kernel k

Then, interpolating is equivalent to condition Y on $Y(x_i) = y_i$ ($i = 1, \dots, n$)

In a Gaussian setting, the conditional process is known explicitly:
 $Y|\{Y(x_i) = y_i, i = 1, \dots, n\}$ is a GP with mean m_c and kernel k_c , with:

$$\begin{aligned} m_c(x) &= m(x) + k(x, X)k(X, X)^{-1}(y - m(X)) \\ k_c(x, x') &= k(x, x') - k(x, X)k(X, X)^{-1}k(X, x') \end{aligned}$$

Notations:

- $y = (y_1, \dots, y_n)^\top$
- $m(X) = (m(x_1), \dots, m(x_n))^\top$
- $k(x, X) = (k(x_1, x), \dots, k(x_n, x))$, $k(X, x) = k(x, X)^\top$
- $k(X, X) = (k(x_i, x_j))_{1 \leq i, j \leq n}$.

¹[Sacks et al., 1989, Rasmussen and Williams, 2006]

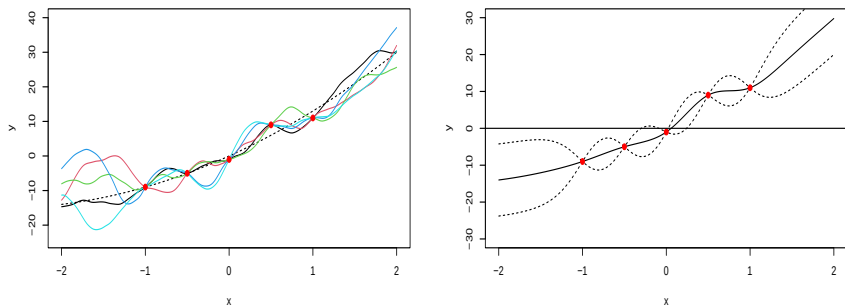


Figure: Example of GP conditional on $\{Y(x_i) = y_i, i = 1, \dots, n\}$ for $n = 5$ points. Left: simulations. Right: mean $m_c(x)$ and 95% prediction intervals $\left[m_c(x) \pm 1.96\sqrt{k_c(x, x)} \right]$.

Outline

1 Introduction

2 Gaussian processes (GP) and GP regression

3 **GPs and linear constraints. Application to linear PDEs**

- Properties
- How to build a linearly constrained GP

4 RKHS

5 Physics-informed GP : non-linear PDEs

Various information... same framework

Question : What is the common point between GPs whose sample paths are...

Centered $\int Y(x) \mu(dx) = 0$

Harmonic $\frac{\partial^2 Y(x)}{\partial x_1^2} + \frac{\partial^2 Y(x)}{\partial x_2^2} = 0$

Symmetric $Y(g.x) = Y(x), \forall g \in G$

Additive $Y(x) = \sum_{j=1}^d Y^j(x_j)$

...

G : Finite group of symmetries.

Various information... same framework

Answer : The information can be reformulated with a **linear operator** L

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$$L(f) = \int f(x) \mu(dx)$$

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Symmetric	$Y(g.x) = Y(x), \forall g \in G$	$L(f)(x) = f(x) - \frac{1}{ G } \sum_{g \in G} f(g.x)$
Additive	$Y(x) = \sum_{j=1}^d Y^j(x_j)$	

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Symmetric $Y(g.x) = Y(x), \forall g \in G$

$$L(f)(x) = f(x) - \frac{1}{|G|} \sum_{g \in G} f(g.x)$$

Additive $Y(x) = \sum_{j=1}^d Y^j(x_j)$

$$L(f)(x) = f(x) - \left(m + \sum_{j=1}^d (E[f(\mathbf{X}) | X_j = x_j] - m) \right)$$

...

with $m = \mathbb{E}[f(\mathbf{X})]$ and $X_1, \dots, X_n$ ind. r.v.

Ref: [Ginsbourger et al., 2016]

Gaussian processes and linear operations

If $Y \sim GP(0, k)$ and L is a linear function acting on the sample paths. of Y , then $LY \sim GP(0, k_L)$ with $k_L(s, t) = L_s L_t k(s, t)$. Here L_s (resp. L_t) means that we apply L on the function $s \mapsto k(s, t)$ (resp. $t \mapsto k(s, t)$).

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The expression of k_L comes, formally, from the bilinearity of covariance:

$$\mathbb{Cov}(LY(s), LY(t)) = L_t(\mathbb{Cov}(LY(s), Y(t))) = L_s L_t \mathbb{Cov}(Y(s), Y(t))$$

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Example if L is the derivative. If Y is a 1D centered GP with (a sufficiently smooth) kernel k , then $(Y'(x))_{x \in \mathbb{X}}$ is a centered GP, with kernel:

$$k_{Y'}(s, t) = \mathbb{Cov}\left(\frac{\partial Y(s)}{\partial s}, \frac{\partial Y(t)}{\partial t}\right) = \frac{\partial}{\partial t} \mathbb{Cov}\left(\frac{\partial Y(s)}{\partial s}, Y(t)\right) = \frac{\partial}{\partial s} \frac{\partial}{\partial t} \mathbb{Cov}(Y(s), Y(t)) = \frac{\partial^2 k}{\partial s \partial t}(s, t)$$

Two results for GPs constrained by linear equalities [Ginsbourger et al., 2016]

- $Y : GP(m, k)$
- L : linear operator on Y such that $L(m) = 0$ (+ integrability condition)

Argumentwise property for kernels

L can be defined in a unique way on the RKHS corr. to k and

$$\forall x : L(Y)(x) = 0 \quad \Leftrightarrow \quad \forall x' : L(k(x', \cdot)) = 0$$

Proof idea: $\text{Cov}(L(Y)(x), Y(x')) = L(\text{Cov}(Y(x), Y(x')))$

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Inheritance to conditional distributions

Let Y_c the GP Y conditional on $Y(x_1), \dots, Y(x_n)$. Then

$$\forall x : L(Y)(x) = 0 \quad \Rightarrow \quad \forall x : L(Y_c)(x) = 0$$

Proof idea: The cond. mean and cond. cov. are linear functions of $k(\cdot, x')$.

PDE constrained GPs [Henderson et al., 2023a]

Given $Y = (Y(z))_{z \in \mathcal{D}}$ a RF, when does $\mathbb{P}(L(Y) = 0) = 1$ hold?

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Given $Y = (Y(z))_{z \in \mathcal{D}}$ a RF, when does $\mathbb{P}(L(Y) = 0) = 1$ hold?

Let $\mathcal{D} \subset \mathbb{R}^d$ be an open set, and $L := \sum_{|\alpha| \leq n} a_\alpha \partial^\alpha$, $a_\alpha \in \mathcal{C}^{|\alpha|}(\mathcal{D})$. Let $Y = (Y(z))_{z \in \mathcal{D}}$ be a second order centered measurable random field, with covariance function k ; assume that $\sigma : x \mapsto k(z, z)^{1/2} \in L^1_{loc}(\mathcal{D})$.

Then the following statements are equivalent.

- $\mathbb{P}(\{\omega \in \Omega : L(Y_\omega) = 0 \text{ in the } \text{distributional sense}\}) = 1$
- $\forall z \in \mathcal{D}, L(k(z, \cdot)) = 0 \text{ in the } \text{distributional sense}.$

This generalises the result of the previous slide to PDE constraints, where solutions are **not always defined pointwise**.

This property is inherited to conditioned GPs (Kriging).

In the distributional sense ?

Sometimes, a function $u : \mathbb{R} \rightarrow \mathbb{R}$ cannot be solution of such a simple equation

$$u'(x) = u_0(x), \quad x \in \mathbb{R}$$

This happens when u_0 is not smooth enough, e.g., $u_0 = \delta_0$.

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This happens when u_0 is not smooth enough, e.g., $u_0 = \delta_0$.

With the theory of **distributions**, we can give a sense to the equation by multiplying with a *test* function ϕ which is C^∞ with compact support:

$$-\langle u, \phi' \rangle = - \int_{\mathbb{R}} u \phi' = \int_{\mathbb{R}} u' \phi = \langle u', \phi \rangle = \langle u_0, \phi \rangle$$

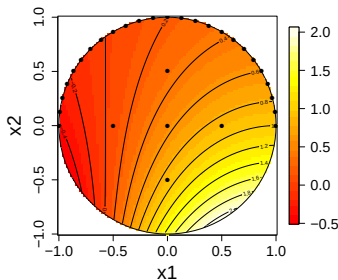
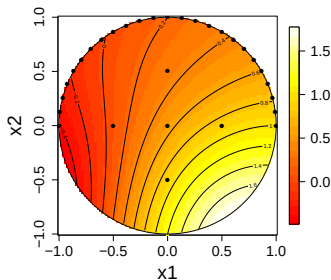
The derivations are reported to ϕ and u itself does not need to be smooth. We replace the **pointwise** sense (for all x) by **local** averages.

Illustration: Maximum of a harmonic function

We compare two GPs for predicting the maximum of a 2D harmonic function (here : $f(\mathbf{x}) = \cos(1 - x_1) \exp(-x_2)$), i.e. solution of $(\partial_{x_1, x_1}^2 + \partial_{x_2, x_2}^2)f = 0$

Gaussian kernel $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(x_1 - x'_1)^2}{\ell_1^2} - \frac{(x_2 - x'_2)^2}{\ell_2^2} \right)$

Harmonic kernel $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(\frac{x_1 x'_1 + x_2 x'_2}{\ell^2} \right) \cos \left(\frac{x_2 x'_1 - x_1 x'_2}{\ell^2} \right)$

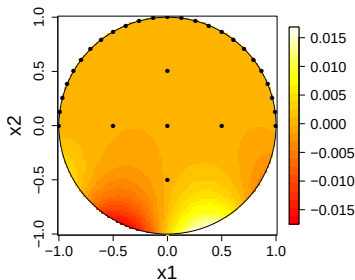
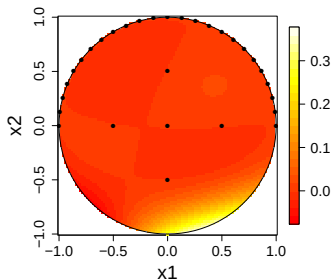


Prediction mean – Left: Gaussian kernel; Right: Harmonic kernel.

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$$\begin{array}{ll} \text{Gaussian kernel} & k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(x_1 - x'_1)^2}{\ell_1^2} - \frac{(x_2 - x'_2)^2}{\ell_2^2} \right) \\ \text{Harmonic kernel} & k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(\frac{x_1 x'_1 + x_2 x'_2}{\ell^2} \right) \cos \left(\frac{x_2 x'_1 - x_1 x'_2}{\ell^2} \right) \end{array}$$

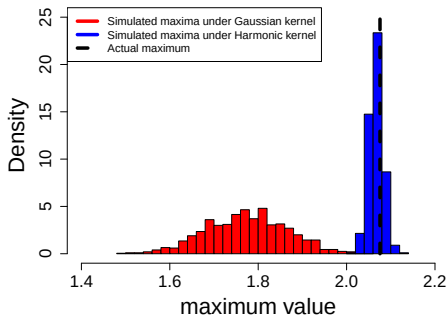


Prediction errors – Left: Gaussian kernel: Right: Harmonic kernel.

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Conditional simulations of the maximum under the two GPs

General receipt

The aim is to construct a GP Y s.t. $(L(Y))(x) = 0$ for all $x \in \mathbb{X}$, where L is a linear operator.

- 1 Find an explicit linear representation of the equation $L(f) = 0$ of the form $f = T(f_0)$, where f_0 can be:
 - ▶ another function (e.g. initial condition, forcing term, ...)
 - ▶ a set of coefficients if we know a basis of solutions of $L(f) = 0$
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- ❹ *Optional but recommended: Choose k_0 s.t. k_f has a closed form!*

Example 1 (elliptic PDE): Harmonic functions in 2 dimensions

The aim is to obtain the kernel proposed by [Schaback, 2009]:

$$k(x, x') = \exp(x_1 x'_1 + x_2 x'_2) \cos(x_2 x'_1 - x_1 x'_2)$$

Here we know a set of solutions in polar coordinates:

$$x \in \mathbb{R}^2 \rightarrow r^n \cos(n\phi), \quad x \in \mathbb{R}^2 \rightarrow r^n \sin(n\phi)$$

and use the convenient expansion

$$Y(r, \phi) = \sum_{n=0}^{+\infty} \frac{1}{\sqrt{n!}} U_n(r, \phi), \quad \text{with} \quad U_n(r, \phi) = r^n \cos(n\phi) A_n + r^n \sin(n\phi) B_n$$

Exercise. Assume that $A_0, B_0, A_1, B_1, \dots$ are i.i.d. $\sim N(0, 1)$.

- Show that $U_n \sim GP(0, k_{U_n})$ with $k_{U_n}((r, \phi), (r', \phi')) = (rr')^n \cos(n(\phi - \phi'))$
- Show that $Y \sim GP(0, k_Y)$ with $k_Y((r, \phi), (r', \phi')) = \exp(rr' \cos(\phi - \phi')) \cos(rr' \sin(\phi - \phi'))$
- Deduce the result.

Example 2 (parabolic PDE): Heat equation

Exercise. Let us consider the heat equation

$$\partial_t u(x, t) - \alpha \partial_{x,x}^2 u(x, t) = 0, \quad u(x, 0) = \phi(x)$$

where $\alpha > 0$ is a **constant**. By applying the Fourier transform, the solution on $\mathbb{R}^2$ is obtained (under suitable conditions) as a spatial convolution:

$$u(x, t) = \int_{\mathbb{R}} G(x - y, t) \phi(y) dy = (G_t * \phi)(x).$$

where $G(x, t) = (4\pi\alpha t)^{-1/2} \exp\left(-\frac{x^2}{4\alpha t}\right) 1_{\mathbb{R}_+}(t)$, and $G_t := G(., t)$.

We now consider that ϕ is unknown, and that $(\phi(x))_{x \in \mathbb{R}}$ is a $\text{GP}(0, k_\phi)$.

- What can you say of the mapping $L : \phi \mapsto \int_{\mathbb{R}} G(\bullet - y, \bullet) \phi(y) dy$?
- Explain why $(u(x, t))_{(x, t) \in \mathbb{R}^2}$ should be a GP on $\mathbb{R}^2$
- Compute formally its mean and its kernel in function of k_ϕ .

Example 2 (parabolic PDE): Heat equation

Partial solution.

$$\begin{aligned}
 \text{Cov}(u(x, t), u(x', t')) &= \text{Cov} \left(\int G_t(x - y) \phi(y) dy, \int G_{t'}(x' - y') \phi(y') dy' \right) \\
 &= \iint \text{Cov} (G_t(x - y) \phi(y), G_{t'}(x' - y') \phi(y')) dy dy' \\
 &= \iint G_t(x - y) G_{t'}(x' - y') \text{Cov} (\phi(y), \phi(y')) dy dy' \\
 &= \iint G_t(x - y) G_{t'}(x' - y') k_\phi(y, y') dy dy'
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We recognize a tensor-product form for G ,

$$k_u((x, t), (x', t')) = \iint [G_t \otimes G_{t'}] \begin{pmatrix} x - y \\ x' - y' \end{pmatrix} k_\phi \begin{pmatrix} y \\ y' \end{pmatrix} dy dy'$$

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This is a 2-dimensional convolution in space:

$$k_u((x, t), (x', t')) = [(G_t \otimes G_{t'}) * k_\phi](x, x')$$

Example 2 (parabolic PDE): Heat equation

For some k_ϕ , the kernel k_u is given explicitly.

For instance, when $k_\phi(y, y') = \exp\left(-\frac{1}{2} \frac{(y-y')^2}{\theta^2}\right)$ is the square exponential kernel, then k_u has the form :

$$k_u\left(\begin{pmatrix} x \\ t \end{pmatrix}, \begin{pmatrix} x' \\ t' \end{pmatrix}\right) = \frac{\sigma_u^2}{\sqrt{2\pi}\sqrt{\theta^2 + 2\alpha(t+t')}} \exp\left(-\frac{1}{2} \frac{(x-x')^2}{\theta^2 + 2\alpha(t+t')}\right).$$

(see, e.g., [[Albert and Rath, 2020](#)])

Linear PDEs with constant coef.: representation with Green functions

Let L be a linear differential operator on $\mathbb{R}^d$, with **constant** coefficients.

The Green 'function' (it can be a distribution!) G is the solution of

$$LG = \delta_0$$

where δ_0 is the Dirac measure.

Then, formally, the solution of $Lu = f$ is represented as

$$u = G * f : x \mapsto \int G(x - y)f(y)dy$$

Indeed,

$$Lu(x) = L(G * f)(x) = \int \underbrace{LG}_{\delta_0}(x - y)f(y)dy = f(x)$$

Linear PDEs with constant coef.: representation with Green functions

Several remarks²

- For the Heat equation, what we called G was indeed the Green function.
- The Green function is not always a function! For the transport equation $\partial_t u + a \partial_x u = 0$ with initial condition $u(x, 0) = u_0$, the Green 'function' is the measure on $\mathbb{R} \times \mathbb{R}$ given by $G = 1_{\mathbb{R}_+}(t) \delta_{at}(dx) dt$
- Numerical bottlenecks: to compute G and $G * f$.

²See e.g. [Henderson, 2023]

Ex 4 - The wave equation [Henderson et al., 2023b]

3D homogeneous wave equation : $\Delta := \partial_{x_1 x_1}^2 + \partial_{x_2 x_2}^2 + \partial_{x_3 x_3}^2$

$$\begin{cases} Lu &= \partial_{tt}^2 u - c^2 \Delta u = \square u = 0, & (x, t) \in \mathbb{R}^3 \times \mathbb{R}^+ \\ u(x, 0) &= u_0(x), \quad \partial_t u(x, 0) = v_0(x). \end{cases} \quad (1)$$

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Representation of u (Kirschhoff) : $F_t = \sigma_{ct}/4\pi c^2 t$ (Green fun.), $\dot{F}_t = \partial_t F_t$

$$u(x, t) = (F_t * v_0)(x) + (\dot{F}_t * u_0)(x). \quad (2)$$

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$$u(x, t) = (F_t * v_0)(x) + (\dot{F}_t * u_0)(x). \quad (2)$$

Assume that u_0, v_0 are unknown $\rightarrow u_0 \sim GP(0, k_{u_0})$ and $v_0 \sim GP(0, k_{v_0})$, independant. u given by (2) is a centered GP, its kernel is

$$k_u((x, t), (x', t')) = [(F_t \otimes F_{t'}) * k_{v_0}](x, x') + [(\dot{F}_t \otimes \dot{F}_{t'}) * k_{u_0}](x, x'). \quad (3)$$

Such **4D convolutions** are not easy to handle.

Fortunately, an **explicit formula** is derived in the **radial symmetry framework**.

More examples of kernels verifying $L(k(z, \cdot)) = 0 \quad \forall z$

Given L , find k_L such that $L(k_L(\cdot, z)) = 0 \quad \forall z$; $\Delta = \sum_{i=1}^d \partial_{x_i}^2$.

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- **Div/Curl free:** $\nabla \cdot u = 0, \nabla \times u = 0$
([Scheuerer and Schlather, 2012]),([Owhadi, 2023])
- **Linear continuum mechanics:** ([Jidling et al., 2018])
- **Helmholtz:** $-\Delta u = \lambda u$ ([Albert and Rath, 2020])
- **(non)stationary Maxwell:** ([Wahlstrom et al., 2013]), ([Jidling et al., 2017]),([Lange-Hegermann, 2018])
- **3D wave equation, transport:** ([Henderson et al., 2023a])
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Often based on representations of solutions of $Lu = 0$ (+BC) of the form

$$u = Gf.$$

Outline

- 1 Introduction
- 2 Gaussian processes (GP) and GP regression
- 3 GPs and linear constraints. Application to linear PDEs
- 4 RKHS**
- 5 Physics-informed GP : non-linear PDEs

RKHS³ (definition)

Reproducing kernel Hilbert spaces (RKHS) are Hilbert spaces of functions $f : \mathbb{X} \rightarrow \mathbb{R}$ such that for all $x \in \mathbb{X}$ the evaluation $f \mapsto f(x)$ is continuous.

This automatically gives a kernel. Indeed, by Riesz theorem, for all $x \in \mathbb{X}$, there exists a unique $k_x \in \mathcal{H}$ s.t. for all $f \in \mathcal{H}$,

$$f(x) = \langle f, k_x \rangle_{\mathcal{H}} \quad (*)$$

Define $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ by $k(x, x') = k_x(x')$. Choosing $f = k_{x'}$ in $(*)$ gives

$$\langle k(x, \cdot), k(x', \cdot) \rangle_{\mathcal{H}} = k(x, x') \quad (**)$$

which shows in particular that k is a kernel.

$(*)$, $(**)$ are called **reproducing properties** and k the **reproducing kernel** of $\mathcal{H}$.

³[Aronszajn, 1950, Berlinet and Thomas-Agnan, 2004, Saitoh and Sawano, 2016]

RKHS (equivalence with psd functions)

We have just seen that if $\mathcal{H}$ is a RKHS, then it provides a kernel k .

Moore-Aronszajn theorem

Conversely, if k is a kernel, one can construct a unique RKHS $\mathcal{H}_k$ with kernel k . This RKHS is given by

$$\mathcal{H}_k = \overline{\text{span}(k(x, \cdot), x \in \mathbb{X})}$$

with inner product defined on the $k(x, \cdot)$'s by $\langle k(x, \cdot), k(y, \cdot) \rangle := k(x, y)$, and extended to $\mathcal{H}_k$ by linearity and continuity.

RKHS (examples)

- Finite-dimensional Hilbert spaces (i.e. Euclidean spaces)
- Infinite dimensional Hilbert spaces with minimal regularity
 - ▶ If $\mathbb{X}$ is a bounded interval, $L^2(\mathbb{X})$ is NOT a RKHS
 - ▶ When $X = \mathbb{R}^d$, the Sobolev space $H^m(\mathbb{X})$ is a RKHS iff $m > d/2$
 $\rightarrow H^1(\mathbb{R}), H^2(\mathbb{R}), \dots$ are RKHS
 - ▶ Matérn kernels correspond to Sobolev spaces with specific norms
[Durrande et al., 2016]

Gaussian processes, splines and RKHS

Correspondence between interpolation spline and GP conditional mean

The interpolation spline is defined by the functional problem

$$(*) \quad \min_{h \in \mathcal{H}} \|h\|_{\mathcal{H}} \quad \text{s.t.} \quad h(x_i) = y_i, \quad i = 1, \dots, n$$

If $\mathcal{H}$ is the RKHS of kernel k , and if $k(X, X) = (k(x_i, x_j))_{1 \leq i, j \leq n}$ is invertible, $(*)$ has a unique solution in the finite dimensional space spanned by the $k(x_i, \cdot)$:

$$\begin{aligned} h^*(x) &= \mathbb{E}[Y(x) | Y(x_i) = y_i, i = 1, \dots, n] \\ &= k(x, X)k(X, X)^{-1}y \end{aligned}$$

where $Y \sim GP(0, k)$, and $\|h^*\|_{\mathcal{H}}^2 = y^\top k(X, X)^{-1}y$.

→ In this sense, GP regression is generalizing interpolation splines.

The first part (reduction to finite dimension) is known as *Representer theorem*, and comes from [Aronszajn, 1950]. The correspondence with GP can be found in [Kimeldorf and Wahba, 1970].

Gaussian processes, splines and RKHS

Link between approximation error & conditional variance [Schaback, 1995]

For all $x \in \mathbb{X}$ and all $h \in \mathcal{H}$ verifying the interpolation constraints $h(X) = y$:

$$|h(x) - h^*(x)| \leq \|h\|_{\mathcal{H}} \times k_c(x, x)^{1/2}$$

N.B., The 'x' part of the upper bound, $k_c(x, x)^{1/2}$, is called *power function*.

Gaussian processes, splines and RKHS

Correspondence between **approximation spline** (kernel ridge regression) and **GP conditional mean for noisy observations**

The **approximation spline** is defined by the **regularized** regression problem

$$(*) \quad \min_{h \in \mathcal{H}} \sum_{i=1}^n (h(x_i) - y_i)^2 + \lambda \|h\|_{\mathcal{H}}^2$$

If $\mathcal{H}$ is the RKHS of kernel k , and if $k(X, X) + \lambda I_n$ is invertible, then $(*)$ has a unique solution in the finite dimensional space spanned by the $k(x_i, \cdot)$:

$$\begin{aligned} h_{\text{opt}}(x) &= \mathbb{E}[Y(x) | Y(x_i) + \varepsilon_i = y_i, i = 1, \dots, n] \\ &= k(x, X)(k(X, X) + \lambda I_n)^{-1} y \end{aligned}$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent $\mathcal{N}(0, \lambda)$, and indep. of the GP Y .

Details for the representer theorem

Consider the minimal norm interpolation problem

$$\min_{h \in \mathcal{H}} \|h\|_{\mathcal{H}}^2, \quad \text{s.t.} \quad y_i = h(x_i), \quad i = 1, \dots, n$$

Let $F = \text{span}(k(x_1, \cdot), \dots, k(x_n, \cdot))$. Write

$$h = f + g, \quad f \in F, g \in F^\perp$$

Now, remark that

$$g(x_i) = \langle g, k(x_i, \cdot) \rangle = 0$$

Thus, the interpolation problem becomes

$$\min_{f \in F, g \in F^\perp} \|f\|_{\mathcal{H}}^2 + \|g\|_{\mathcal{H}}^2, \quad \text{s.t.} \quad y_i = f(x_i), \quad i = 1, \dots, n$$

which implies $g = 0$.

Details for the representer theorem: extensions

Consider a linear operator $L : \mathcal{H} \rightarrow \mathbb{X}, \mathbb{R}$ and the problem

$$\min_{h \in \mathcal{H}} \|h\|_{\mathcal{H}}^2, \quad \text{s.t.} \quad y_i = (Lh)(x_i), \quad i = 1, \dots, n$$

Assume that for all $x \in \mathbb{X}$ the linear form $h \in \mathcal{H} \mapsto Lh(x)$ is continuous. Then by Riesz theorem,

$$\exists! k_{L,x} \in \mathcal{H}, \text{ s.t. } Lh(x) = \langle h, k_{L,x} \rangle$$

Let $F = \text{span}(k_{L,x_1}, \dots, k_{L,x_n})$. Write

$$h = f + g, \quad f \in F, g \in F^\perp$$

Now, remark that

$$(Lg)(x_i) = \langle g, k_{L,x_i} \rangle = 0$$

The end of the proof is the same.

Extension of the representer theorem to derivatives

Let O be an open set in $\mathbb{R}$. Let us denote $\partial_x k(u, v)$ (resp. $\partial_y k(u, v)$) the derivative of k w.r.t. the 1st variable (resp. the 2nd), evaluated at (u, v) .

Representation of derivative in RKHS [Saitoh and Sawano, 2016]

Assume that $k, \partial_x k, \partial_y k, \partial_{xy}^2 k$ (and $\partial_{yx}^2 k$) exist and are continuous on $O \times O$. Then, $\partial_x k(x, \cdot) \in \mathcal{H}$ (for all $x \in O$), all $f \in \mathcal{H}$ are of class C^1 in O and

$$f'(x) = \langle f, \partial_x k(x, \cdot) \rangle \quad (\forall x \in O)$$

Furthermore, for all $x, x' \in O$,

$$\begin{aligned} \langle k(x, \cdot), \partial_x k(x', \cdot) \rangle &= \partial_y k(x, x') = \partial_x k(x', x) \\ \langle \partial_x k(x, \cdot), \partial_x k(x', \cdot) \rangle &= \partial_{xy}^2 k(x, x') = \partial_{yx}^2 k(x', x) \end{aligned}$$

- Mind that f' may not always belong to $\mathcal{H}$. Can you see why?
- In general, if $L = \sum_{|\alpha| \leq N} a_i \partial^\alpha$ is a linear diff. operator on $\mathbb{R}^d$, and if k is smooth enough, then L is continuous and $Lu(x) = \langle u, k_{L,x} \rangle$ with $k_{L,x} \in \mathcal{H}$; the representer theorem applies [Wendland, 2004].

Extension of the representer theorem to derivatives

Representer theorem with derivatives

Under the assumptions of the previous slide, the minimization problem

$$(*) \quad \min_{h \in \mathcal{H}} \|h\|_{\mathcal{H}} \quad \text{s.t.} \quad h(x_i) = y_i, \quad h'(x_i) = d_i \quad (i = 1, \dots, n)$$

can be solved in the finite-dim. space spanned by the $k(x_i, \cdot)$ and the $\partial_x k(x_i, \cdot)$

Furthermore, if $\begin{bmatrix} k(X, X) & \partial_y k(X, X) \\ \partial_y k(X, X)^\top & \partial_{xy}^2 k(X, X) \end{bmatrix}$ is invertible, then $(*)$ has a unique solution given by

$$h^*(x) = \begin{bmatrix} k(x, X) & \partial_y k(x, X) \end{bmatrix} \begin{bmatrix} k(X, X) & \partial_y k(X, X) \\ \partial_y k(X, X)^\top & \partial_{xy}^2 k(X, X) \end{bmatrix}^{-1} \begin{bmatrix} y \\ d \end{bmatrix}$$

Link with GP regression with derivatives

If k is smooth enough, the joint process (Y, Y') is well-defined and is a GP.

Exercise. Using the formula of Gaussian conditioning, show that the conditional process $Y|\{Y(X) = y, Y'(X) = d\}$ is a GP with mean m_c^{der} and kernel k_c^{der} :

$$\begin{aligned} m_c^{\text{der}}(x) &= h^*(x) \\ k_c^{\text{der}}(x, x') &= k(x, x') \\ &\quad - \begin{bmatrix} k(x, X) & \partial_y k(x, X) \end{bmatrix} \begin{bmatrix} k(X, X) & \partial_y k(X, X) \\ \partial_y k(X, X)^\top & \partial_{xy}^2 k(X, X) \end{bmatrix}^{-1} \begin{bmatrix} k(x', X) \\ \partial_y k(x', X) \end{bmatrix} \end{aligned}$$

Explain to what correspond each of the terms: $\partial_y k(x, X)$, $\partial_{xy}^2 k(x, X)$, ...

Outline

- 1 Introduction
- 2 Gaussian processes (GP) and GP regression
- 3 GPs and linear constraints. Application to linear PDEs
- 4 RKHS
- 5 **Physics-informed GP : non-linear PDEs**

Solving non-linear PDEs with GPs

The idea is to use the functional view of GPs, i.e. RKHS.

The next slides are based on [[Chen et al., 2021](#)].

Solving non-linear PDEs with GPs

As an example, we focus on the PDE

$$\begin{aligned} -\Delta u(x) + (u(x))^3 &= f(x) & \forall x \in \Omega \\ u(x) &= g(x) & \forall x \in \partial\Omega \end{aligned}$$

where Ω is an open set of $\mathbb{R}^d$ and $f : \Omega \rightarrow \mathbb{R}$, $g : \partial\Omega \rightarrow \mathbb{R}$ are continuous functions.

The aim is to predict $u(x)$ knowing values of f on Ω and values of g on $\partial\Omega$.

Solving non-linear PDEs with GPs

Let $\mathcal{H}$ be a RKHS with kernel k .

Let us consider the minimization problem

$$\min_{u \in \mathcal{H}} \|u\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad \begin{cases} -\Delta u(x_m) + (u(x_m))^3 = f(x_m) & m \in \mathcal{C}_{\Omega} = \{1, \dots, M_{\Omega}\} \\ u(x_m) = g(x_m) & m \in \mathcal{C}_{\partial\Omega} = \{M_{\Omega} + 1, \dots, M\} \end{cases}$$

where $x_1, \dots, x_M$ are fixed points (collocation points).

As the PDE is non-linear, we cannot use the representer theorem. The trick is to recast this problem as the bilevel optimization problem (denote $\mathcal{C} = \mathcal{C}_{\Omega} \cup \mathcal{C}_{\partial\Omega}$):

$$\min_{(z_m^{(1)}, z_m^{(2)})_m} \left(\min_{\substack{u \in \mathcal{H} \\ u(x_m) = z_m^{(1)}, m \in \mathcal{C} \\ \Delta u(x_m) = z_m^{(2)}, m \in \mathcal{C}_{\Omega}}} \|u\|_{\mathcal{H}}^2 \right) \quad \text{s.t.} \quad \begin{cases} -z_m^{(2)} + (z_m^{(1)})^3 = f(x_m) & m \in \mathcal{C}_{\Omega} \\ z_m^{(1)} = g(x_m) & m \in \mathcal{C}_{\partial\Omega} \end{cases}$$

Solving non-linear PDEs with GPs

Inner optimization

$$\min_{u \in \mathcal{H}} \|u\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad \begin{cases} u(x_m) = z_m^{(1)} & m \in \mathcal{C} \\ \Delta u(x_m) = z_m^{(2)} & m \in \mathcal{C}_\Omega \end{cases}$$

where all $z_m^{(1)}, z_m^{(2)}$ are real numbers.

As Δ is a linear operator, we can use a version of the representer theorem. Let $Y \sim GP(0, k)$. The solution is unique and given by

$$u_z^*(x) = \mathbb{E} \left(Y(x) | \{Y(x_m) = z_m^{(1)}\}_{m \in \mathcal{C}}, \{\Delta Y(x_m) = z_m^{(2)}\}_{m \in \mathcal{C}_\Omega} \right)$$

and the value at minimum is quadratic w.r.t. $z = (z_1^{(1)}, \dots, z_M^{(1)}, z_1^{(2)}, \dots, z_{M_\Omega}^{(2)})$

$$\|u_z^*\|_{\mathcal{H}}^2 = z^\top K(X, X)^{-1} z$$

where $K(X, X) = \text{Cov}(Y(x_1), \dots, Y(x_M), \Delta Y(x_1), \dots, \Delta Y(x_{M_\Omega}))$.

Solving non-linear PDEs with GPs

Outer minimization problem

Step 2. Minimize $\|u_z^*\|_{\mathcal{H}}^2 = z^\top K(X, X)^{-1} z$ w.r.t $(z_m^{(1)}, z_m^{(2)})_{m \in \mathcal{C}_\Omega} \in \mathbb{R}^{2|\mathcal{C}_\Omega|}$ subject to non-linear constraints inherited from the PDE

$$\min_{(z_m^{(1)}, z_m^{(2)})_{m \in \mathcal{C}_\Omega}} z^\top K(X, X)^{-1} z \quad \text{s.t.} \quad \begin{cases} -z_m^{(2)} + (z_m^{(1)})^3 = f(x_m) & m \in \mathcal{C}_\Omega \\ z_m^{(1)} = g(x_m) & m \in \mathcal{C}_{\partial\Omega} \end{cases}$$

This is a **finite-dimensional quadratic optimization problem**.

The authors prove the convergence of the solution of this problem to the solution u of the PDE when the number of collocation points M goes to infinity, *under the assumption that $u \in \mathcal{H}$ (and other technical ones)*
 $\rightarrow$ *requires to choose the kernel k such that $u \in \mathcal{H}$*

Solving non-linear PDEs with GPs

Notice that **the constraints can often be removed** by substitution.

$$\min_{(z_m^{(1)}, z_m^{(2)})_{m \in \mathcal{C}_\Omega}} z^\top K(X, X)^{-1} z \quad \text{s.t.} \quad \begin{cases} -z_m^{(2)} + (z_m^{(1)})^3 = f(x_m) & m \in \mathcal{C}_\Omega \\ z_m^{(1)} = g(x_m) & m \in \mathcal{C}_{\partial\Omega} \end{cases}$$

$\Downarrow$ rewrite $z_m^{(2)}$ in function of $z_m^{(1)}$

$$\min_{(z_m^{(1)})_{m \in \mathcal{C}_\Omega}} v^\top K(X, X)^{-1} v \quad \text{with} \quad v = \begin{bmatrix} (z_m^{(1)})_{m \in \mathcal{C}_\Omega} \\ (g(x_m))_{m \in \mathcal{C}_{\partial\Omega}} \\ ((z_m^{(1)})^3 - f(x_m))_{m \in \mathcal{C}_\Omega} \end{bmatrix}$$

The outer problem is an unconstrained non-linear optimization problem.

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